BBP

RASD & DD Presentation

Alessandro Burato, Gabriele Crimella, Davide Faccini

Politecnico di Milano

January 17, 2026



POLITECNICO
MILANO 1863

Outline

1. RASD
   - Introduction
   - Overall Description
   - Specific Requirements
   - Alloy
2. DD
   - Architectural Design
   - Implementation, Integration and Test Plan
3. References

1 RASD
- Introduction
- Overall Description
- Specific Requirements
- Alloy

2 DD

3 References

1 RASD
Introduction
Overall Description
Specific Requirements
Alloy

2 DD

3 References

Introduction

- Key goals (G1-G14):
  - **Data Acquisition**: Automated tracking & manual reporting (G10, G11).
  - **Data Quality**: Merging user contributions (G8) & Anomaly detection (G9).
  - **Value**: Path optimality scoring based on road status (G5).

Phenomena

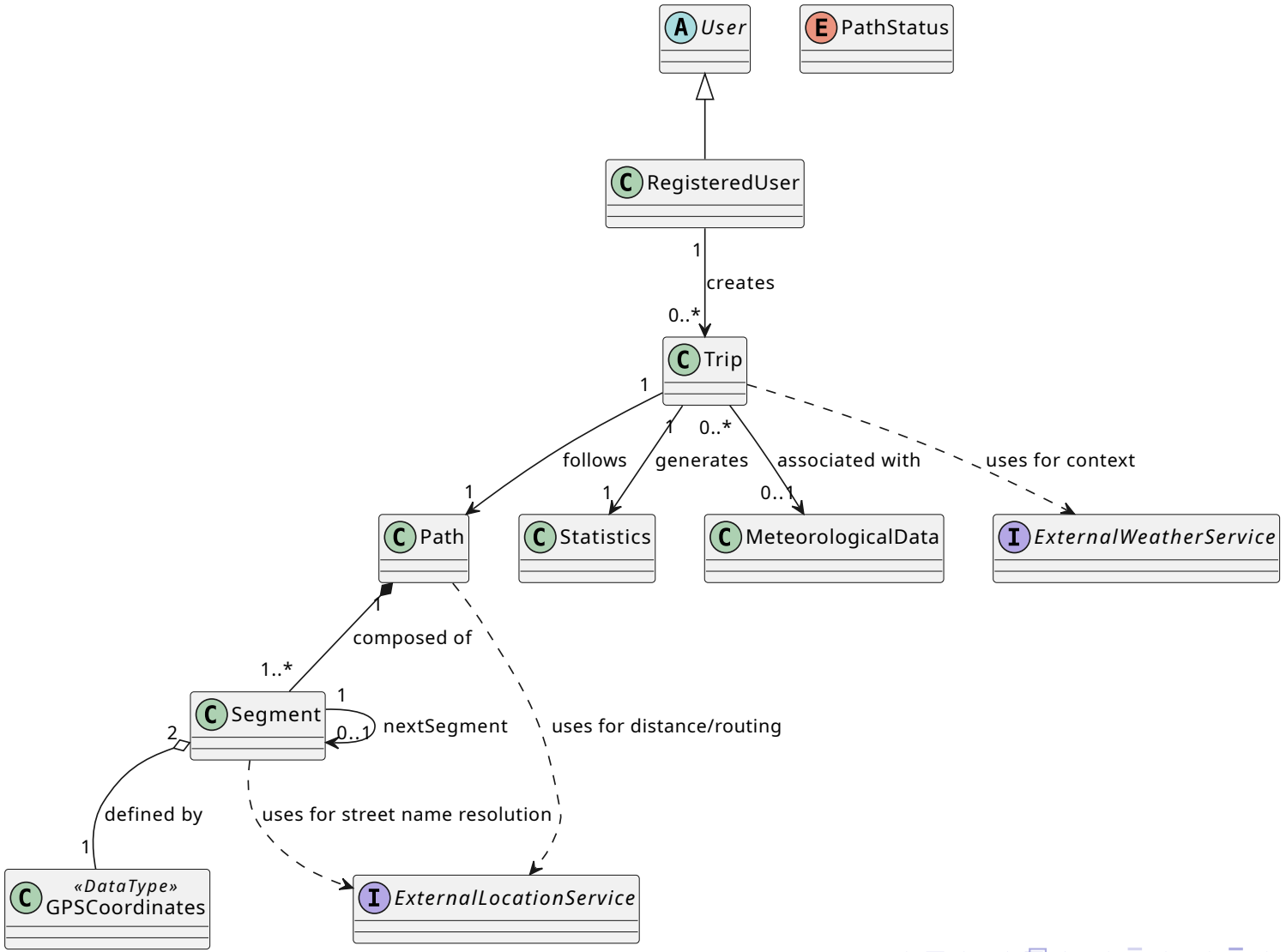
- World Phenomena (WP):
  - *Physical movement*
  - *Road status changes (potholes)*
  - *Weather conditions*
- Shared Phenomena (SP):
  - World-controlled:
    - *Sensors transmitting*
    - *User inputs*
  - Machine-controlled:
    - *Map visualization*
    - *Confirmation requests*

1 RASD
- Introduction
- Overall Description
- Specific Requirements
- Alloy

2 DD

3 References

Class diagram

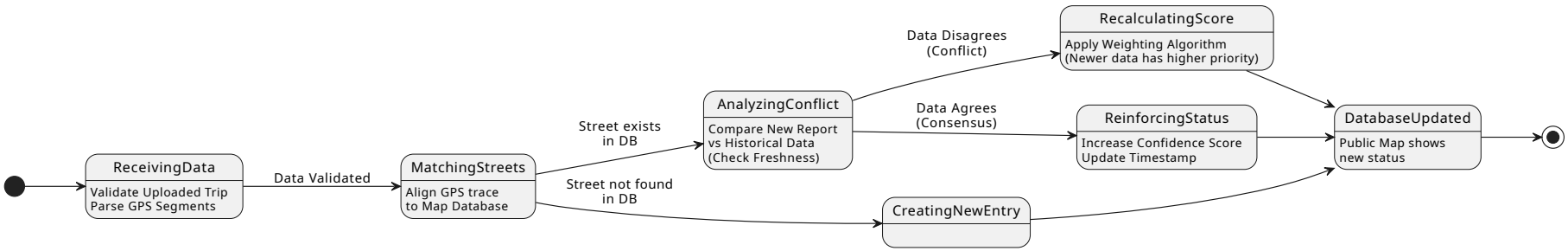


System Dynamics & Data Lifecycle

- **Acquisition Strategy**: hybrid model using **inference** (speed thresholds) and **topological reconstruction** (manual)
- **Validation Pattern**: sensor anomalies are flagged as "**candidates**" to filter **false positives** via user confirmation
- **Consistency Logic**: server-side **weighting algorithm** resolves conflicts based on data freshness and consensus

State Diagram Example

Path Aggregation

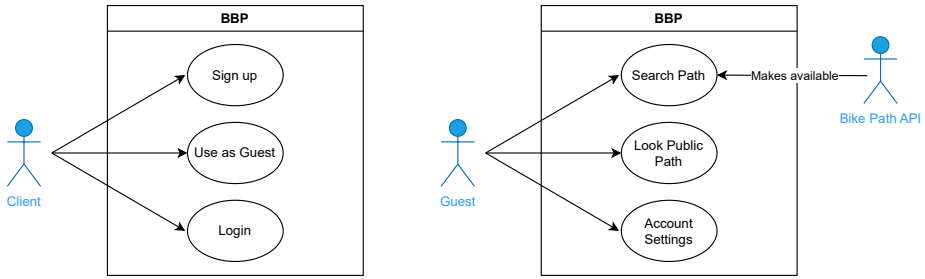


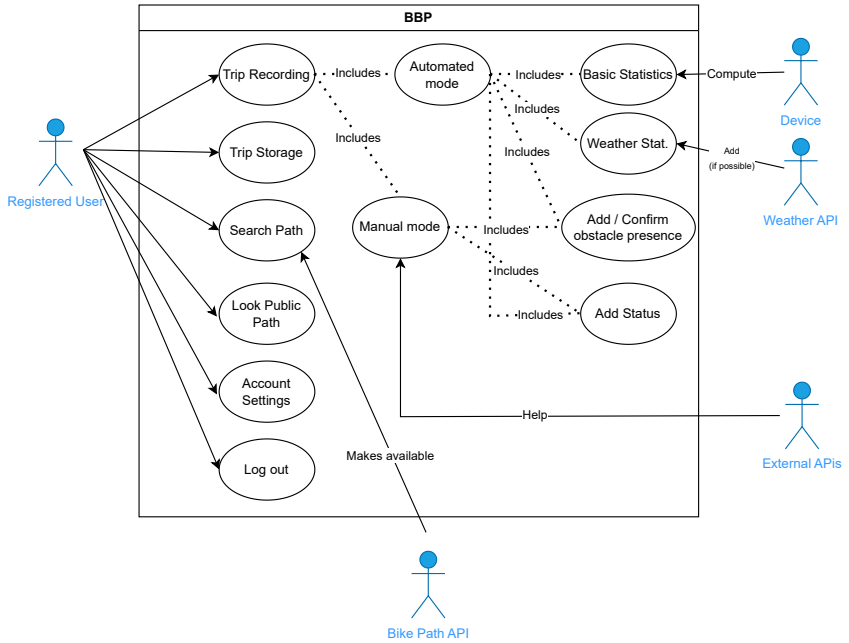
1 RASD
- Introduction
- Overall Description
- Specific Requirements
- Alloy

2 DD

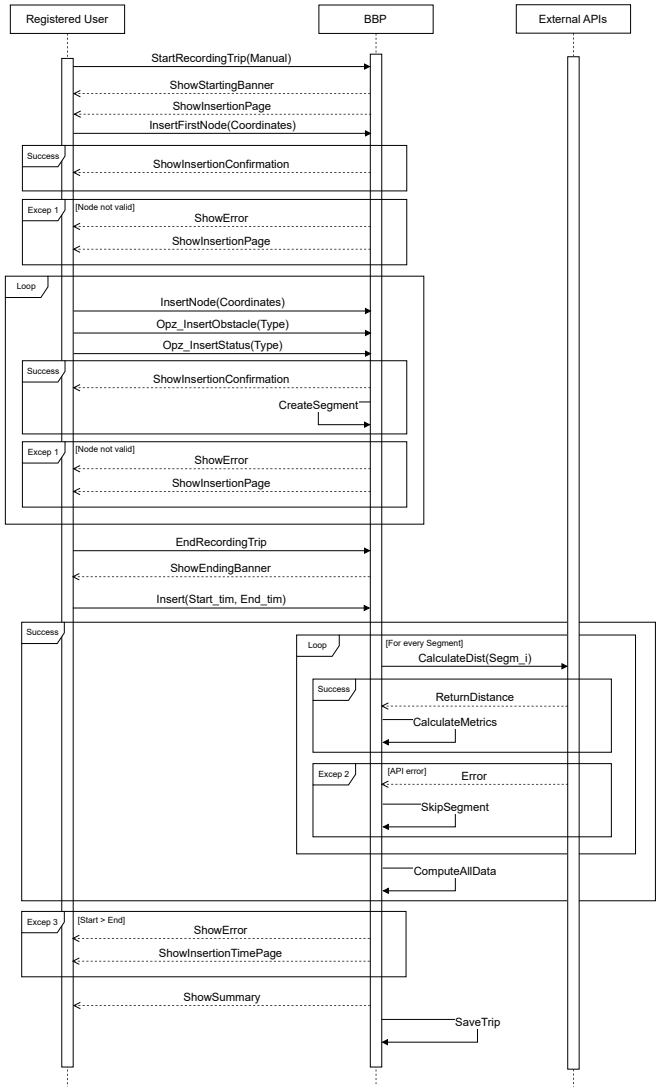
3 References

Specific Requirements





Specific Requirements



Specific Requirements

- **Realibility**
- **Availability**
- **Security**
- **Maintainability**
- **Portability**

1 RASD
- Introduction
- Overall Description
- Specific Requirements
- Alloy

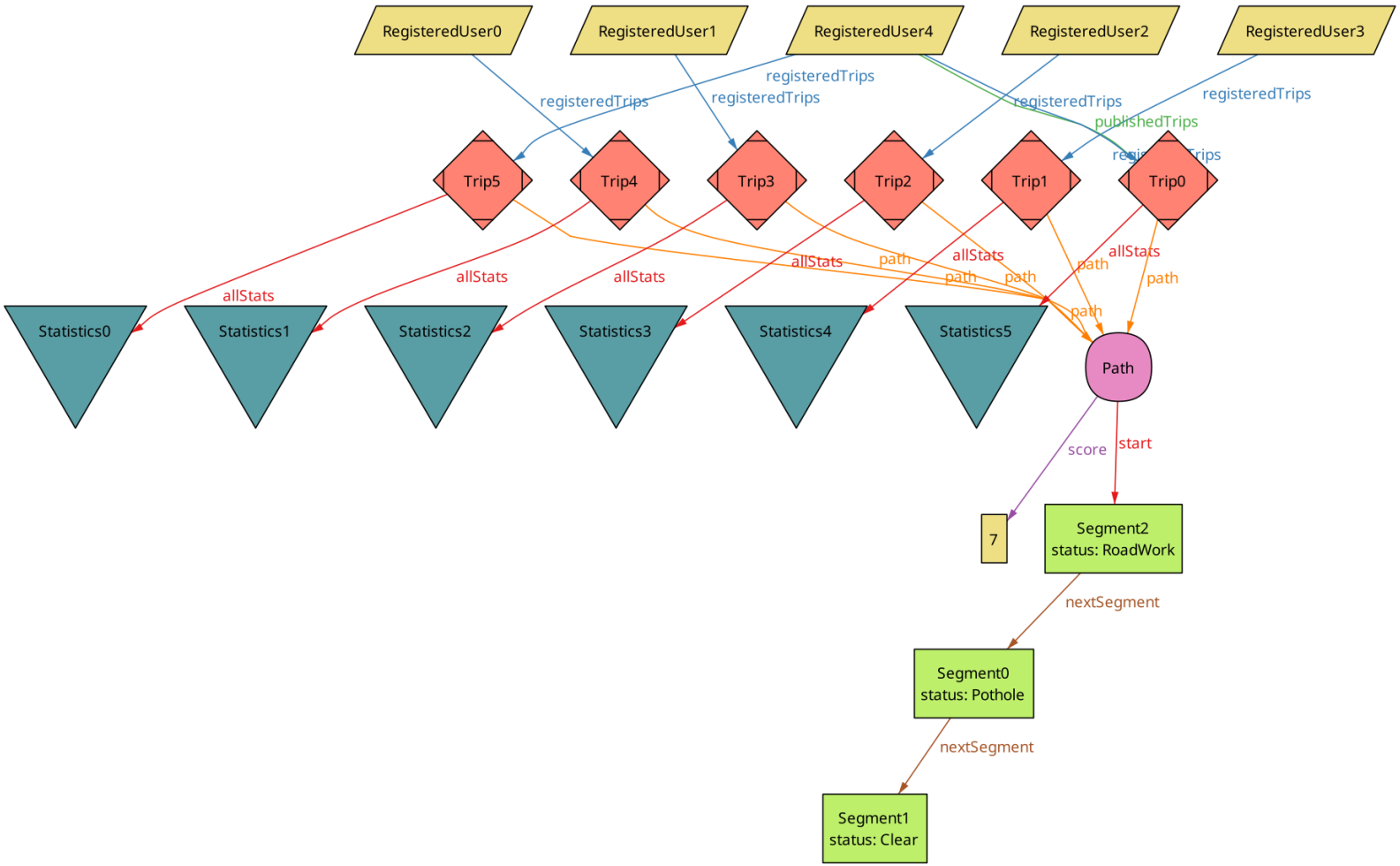
2 DD

3 References

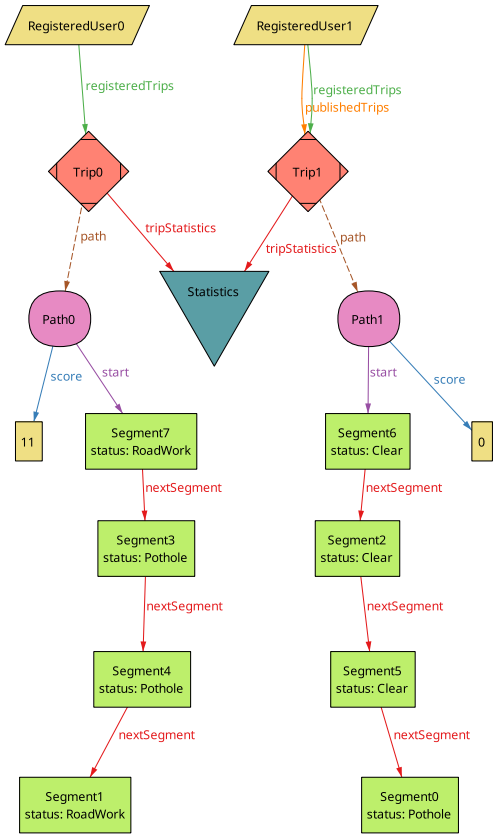
Alloy

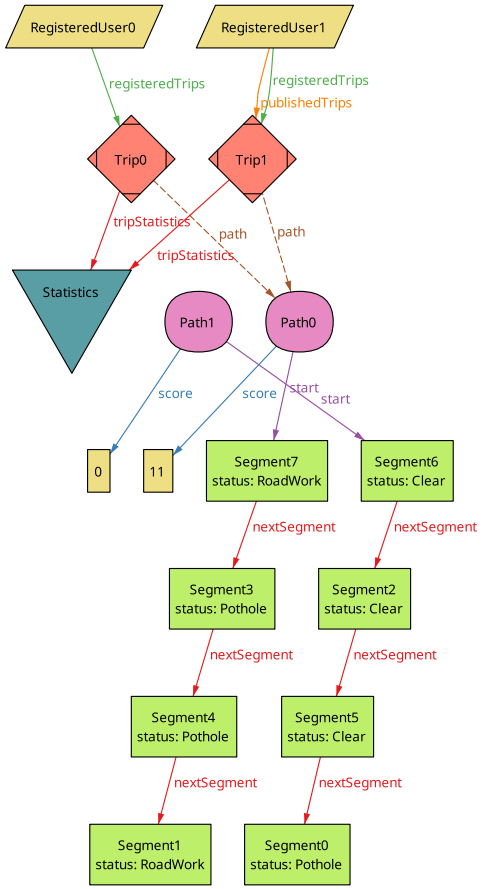
- **Static model analysis**
- **Merging of two paths**

Static Model Analysis



Merging of Two Paths





1 RASD

2 DD
Architectural Design
Implementation, Integration and Test Plan

3 References

1 RASD

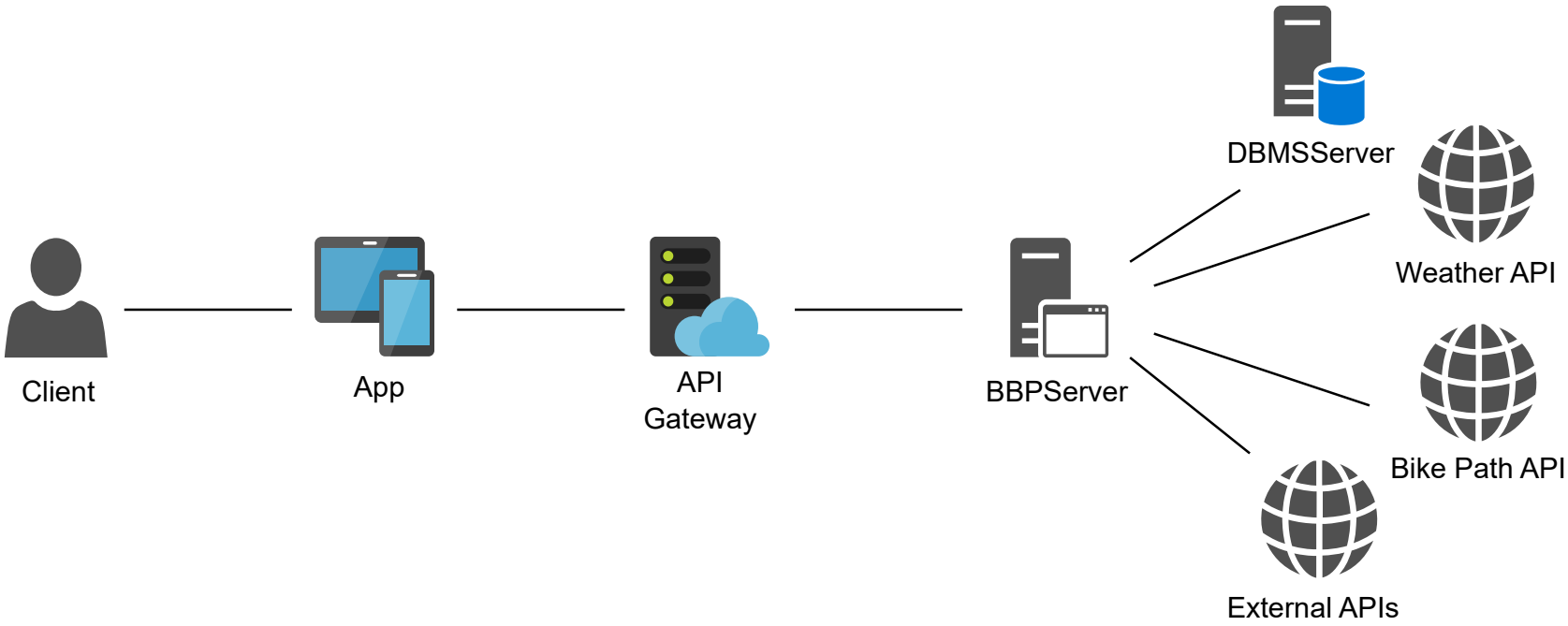
2 DD

Architectural Design

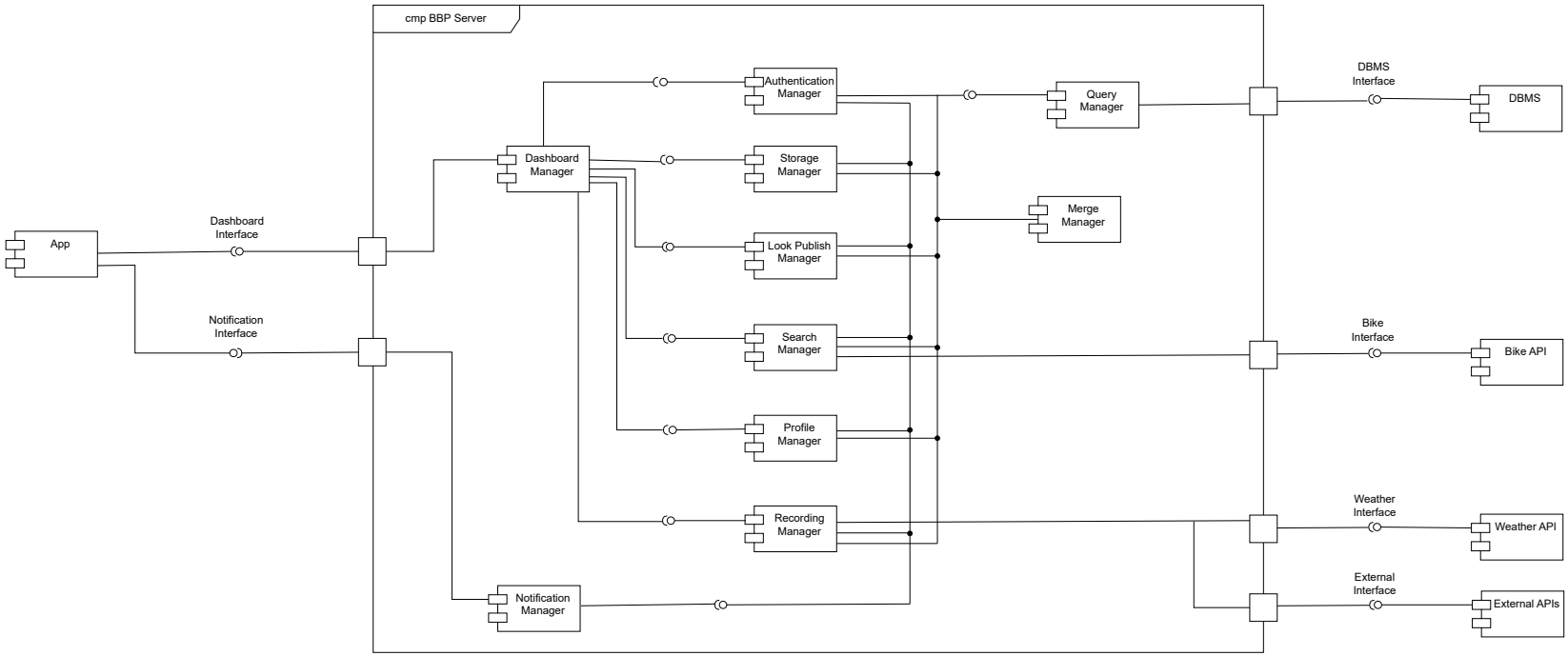
Implementation, Integration and Test Plan

3 References

Architectural Design



Architectural Design



1 RASD

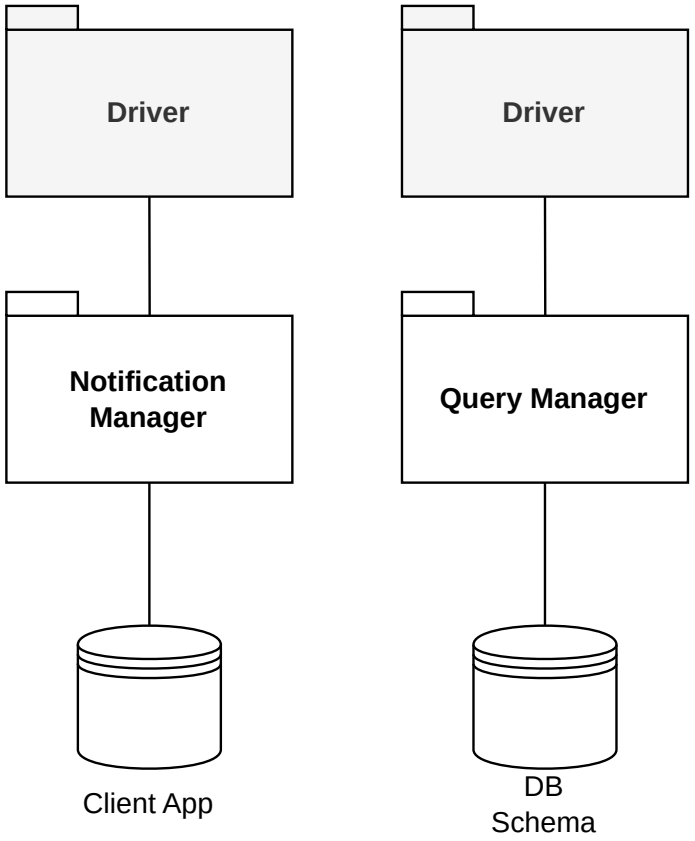
2 DD

Architectural Design

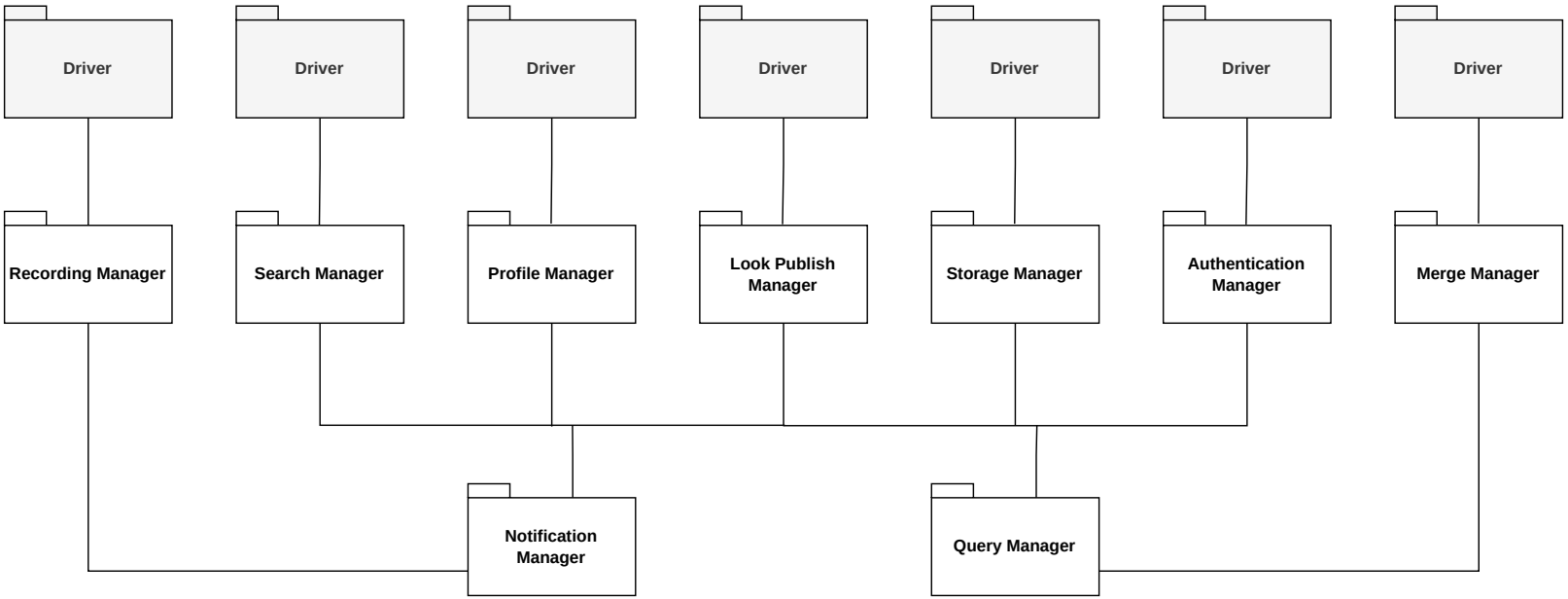
Implementation, Integration and Test Plan

3 References

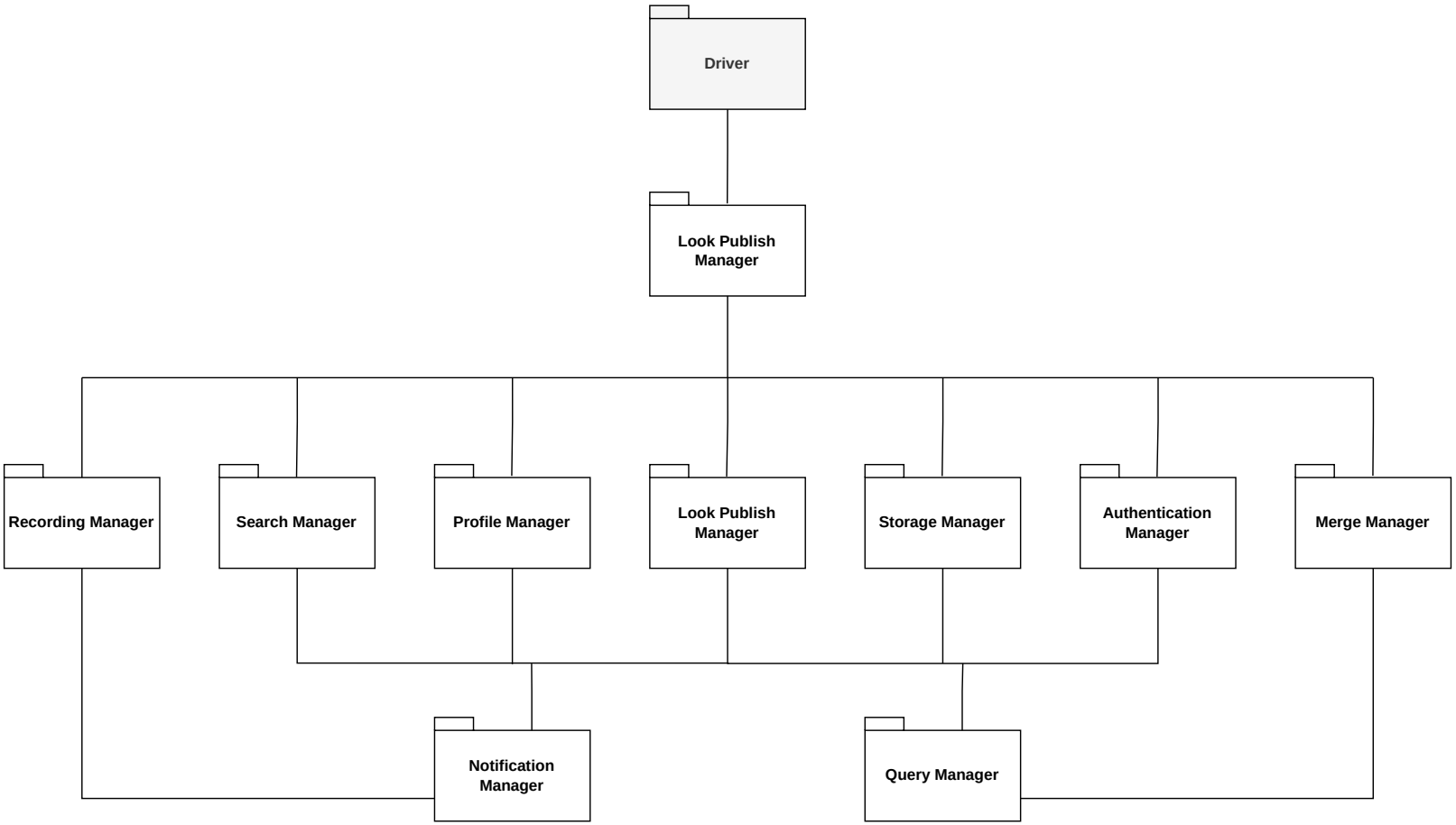
Query and Notification Manager



Parallel Integration



Dashboard Manager



System Testing Strategy

- **Functional Testing**
- **Load Testing**
- **Performance Testing**
- **Stress Testing**
- **User Interface Testing**

1 RASD

2 DD

3 References

References

[1] Author, *Title of the Paper*, Journal, Year.

Thank you!